

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Chemistry
2. Subject Name: Quantum Chemistry and Atomic Spectroscopy
L-T-P: 3-1-0 Credits: 4
3. Pre-requisites: NIL
4. Syllabus and reference books:

Syllabus:

Module 1: Atomic and molecular structure:

- a) Review of quantum mechanics: Schrodinger wave equation, Eigenfunction, eigenvalues and operators, Interpretation of wave functions
- b) Exactly solvable models: Free particle, Particle-in-a-box, Rigid rotor and angular momentum, Harmonic oscillator, Hydrogen atom, Spin
- c) Introduction to self-consistent field theories
- d) Valence Bond and Molecular Orbital theories : Hydrogen molecule ion

Module 2: Atomic spectroscopy:

- a) Historical development: The Lyman, Balmer, Paschen, Brackett, and Pfund series of H-atom, The Ritz combination principle, The Ritz formula, The Hicks formula, Series formula applied to alkali metals, Normal and abnormal series, Hydrogen and Pickering series
- b) Fine structure and spinning electron: doublet fine structure in alkali metals and Boron group of Elements, spin-orbit interactions, Zeeman effect, Stark effect

Module 3: Intermolecular interaction:

- a) Electrical properties of molecules including permanent and induced dipoles,
- b) Dispersion interaction;
- c) Empirical models of intermolecular interaction.

Reference Books:

- 1) Physical Chemistry: A Molecular Approach by D.A.McQuarrie and J.D. Simon
(Publisher: University Science Books, 1997)

- 2) Quantum Chemistry by Ira N. Levine, 7th Ed, (Publisher: Pearson Education India, 2016)
- 3) Introductory Quantum Chemistry by A. K. Chandra, 4th Ed, Tata McGraw-Hill, 1994
- 4) PHYSICAL CHEMISTRY, Volume – IV : QUANTUM MECHANICS by Nabakumar Bera, Subhasree Ghosh, Paulami Ghosh, 1st Ed, Techno India, 2022
- 5) Introduction to Atomic Spectra by White by McGraw-Hill Book Company, 1934

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Review of quantum mechanics: Schrodinger wave equation, Eigenfunction, eigenvalues and operators, Interpretation of wave functions	04
2.	Exactly solvable models: Free particle, Particle-in-a-box, Rigid rotor and angular momentum, Harmonic oscillator, Hydrogen atom, Spin	18
3.	Introduction to self-consistent field theories	02
4.	Valence Bond and Molecular Orbital theories : Hydrogen molecule ion	02
5.	Atomic spectroscopy	10
7.	Intermolecular interaction	04
Total number of hours		40